

Optimising CO2 Sequestration Networks

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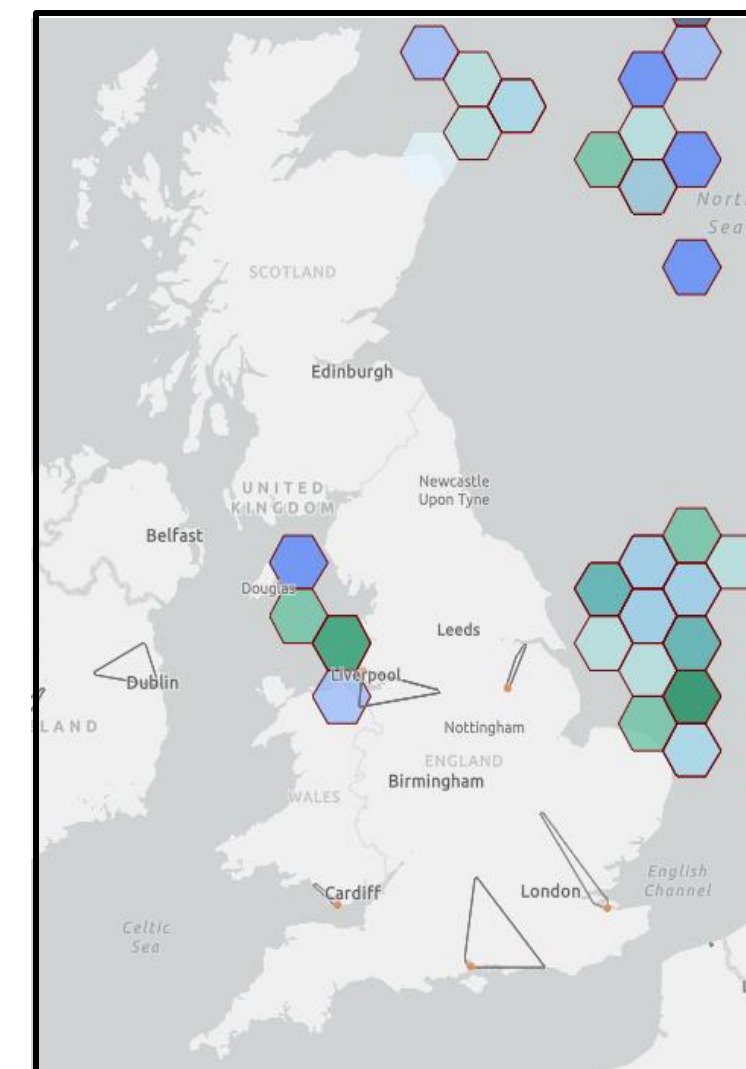
Project Overview

In this project we explore how CO2 sequestration networks should be structured to minimise cost, accelerating progress to net zero.

Study of hubs and CO2 sequestration networks is not new. Infact, some tools already exist. These include OGCI's interactive CCUS hub tool. This tool allows user to explore the impact on the cost per tn CO2 sequestered as hub locations and sequestration sites are changed.

Our work looks beyond this tool by analysing new possible hub locations and forming an effective pipeline network. Our project contained 3 key stages:

- Research – Learning about CO2 sequestration's recent developments and challenges
- Data analysis – Distilling our raw data sources to a relevant subset of point sources and sequestration sites
- Algorithm Development – Using our mathematical data modelling skills to build an effective algorithm



(CCUS HUB Interactive Map)



Transporting CO2



Gas Pipeline

Sequestration networks can be constructed with various methods of transport, including pipelines, tanker trucks and trains. In all these cases CO2 is liquified to reduce volume. Previous studies have found motorway transportation is more optimal when point sources are close together (where infrastructure costs out way benefit) . Train and pipelines being the options with greater distance (Stolaroff, Joshua et al 2021).

Hubs are key to reduce unnecessary transport routes between point sources and sequestration sites.

Offshore hubs can offer the same benefits allowing management of CO2 when sinks become saturated or the network changes. The onshore network will have a more complicated structure as motorway and rail networks are integrated alongside pipelines. Also, onshore networks need to avoid protected areas such as national parks and urban areas.



UK Motorway Network

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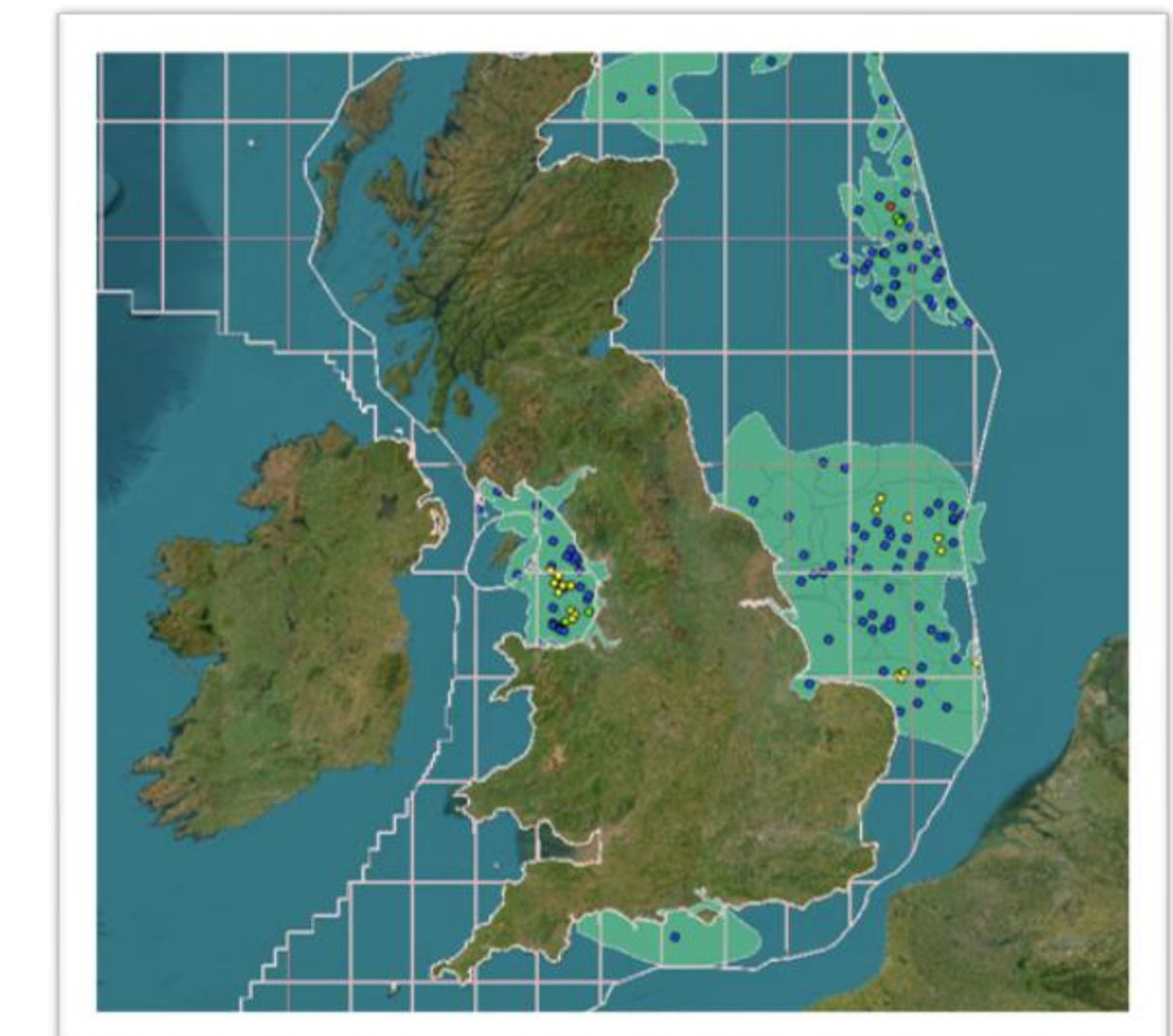
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Investigating Sequestration Sites

- Sequestration sites are potential CO2 storage deposits located in the sea
- Triassic period of formation of these sites have lower seal degradation likelihood
- Sandstone is mineralogically and geochemically suitable as CO2 storage
- Saline aquifers are the most promising sites for storage as they are not limited by their reservoir size unlike other site types
- The Ormskirk (East Irish Sea) and Bunter (Southern North Sea) sandstone formations demonstrate all the required characteristics for CO2 storage including closed structures, high porosity, permeability, and are overlain by an effective seal
- We aim to store England's CO2 in these sequestration sites

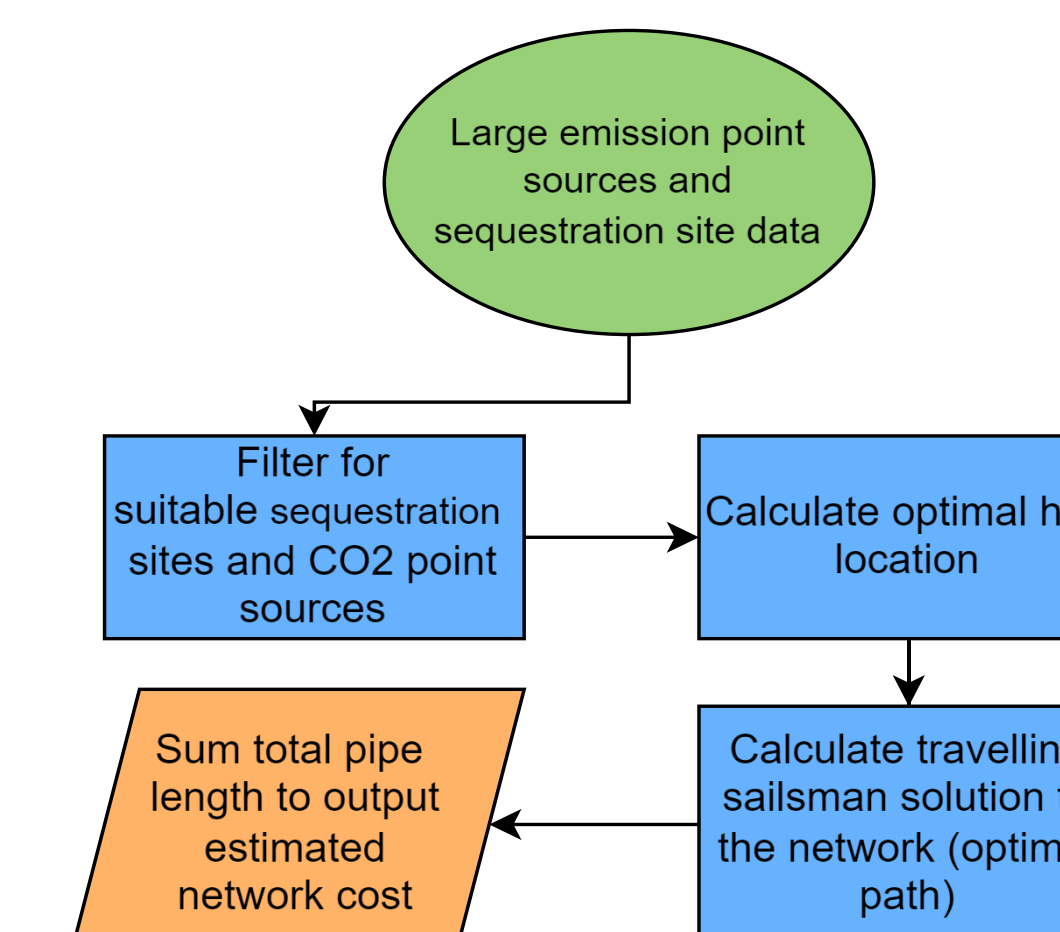


Powerstation Locations

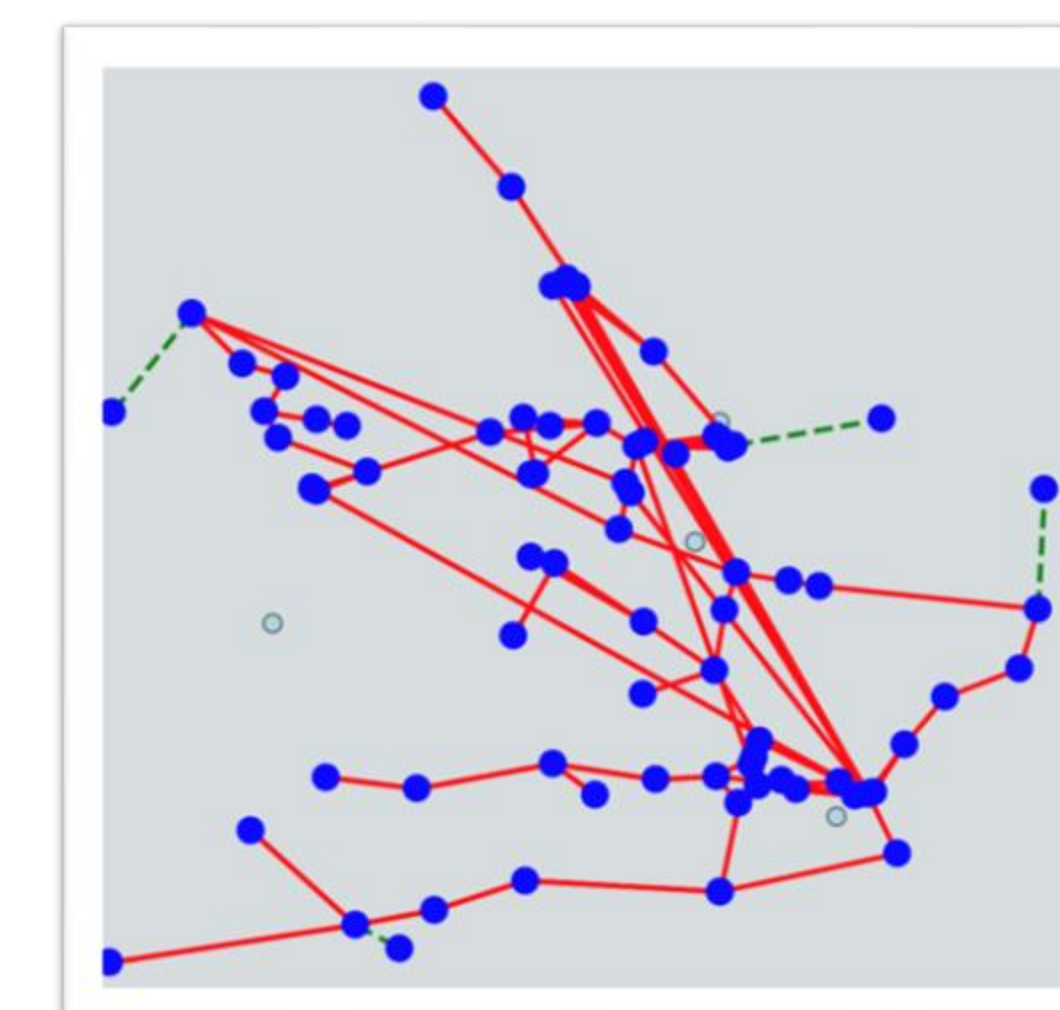


Potential Sequestration Sites

Method and Results



Flowchart of Algorithm



Network map produced by Algorithm

In the figure below, we have mapped the optimum CO2 pipeline network for England. Our algorithm uses a combination of...

- A solution to the travelling salesman problem
- The haversine equation
- Shortest distance calculations

The network connects our selected sequestration sites to carbon hubs with greatest price efficiency. The haversine equation and the shortest distance are used to established hubs which link to sinks.

With the data obtained from CO2 stored and NAEI, and after applying our algorithm. The network metrics are:

- Total CO2 generated: 180Mt/yr
- Potential storage capacity: 1900Mt
- Pipeline Length: 5500km
- Approximate Pipeline Cost: £2.75 billion

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